

Sec 4.6 – Rates and Related Rates

(these notes are adapted from a variety of sources, including notes provided by Dr. Kristin Camenga)

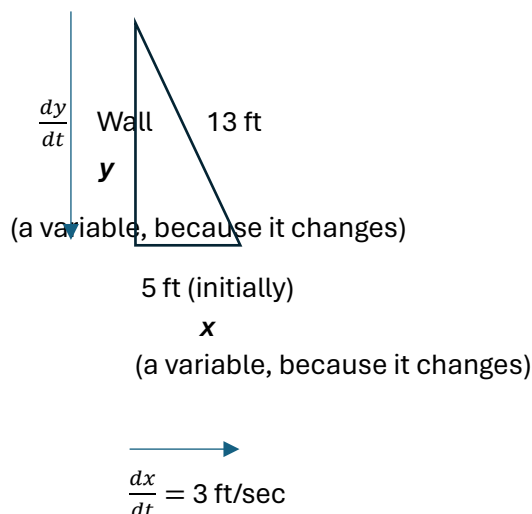
You may have noticed that we skipped forward. Last class we covered Sec 3.7 – Implicit Differentiation, and today we will cover Sec 4.6 – Rates and Related rates.

This is intentional. These Related Rates problems are applications of Implicit Differentiation. We need Implicit Differentiation to solve them. Note that there are generally two parts: (1) setting up the problem and then (2) executing the Implicit Differentiation to solve it.

Let's dive right in with our first problem. Here's a link to a similar simulation: <https://www.geogebra.org/m/Yfudjzga>

Problem 1: A 13-ft ladder leans against a wall. The bottom of the ladder is 5 feet from the wall and slides away from the wall at a rate of 3 ft/s. How fast is the top of the ladder moving down the wall?

Draw a picture of this situation. What parts of your picture are changing with time?



Now, write an equation relating the variables in the diagram.

$$x^2 + y^2 = 13^2$$

Note that both x and y are functions of time, t .

And initially, what do we know about x , y , $\frac{dx}{dt}$, $\frac{dy}{dt}$?

$x = 5$ and $\frac{dx}{dt} = 3 \text{ ft/sec}$ are given, and using the Pythagorean Theorem, initially $5^2 + y^2 = 13^2$, which gives $y = 12$.

We know three of the four quantities, x , y , $\frac{dx}{dt}$, and we are asked for $\frac{dy}{dt}$.

Since x and y are functions of time, t , when we differentiate the equation $x^2 + y^2 = 13^2$ with respect to time we will use implicit differentiation:

$$\frac{d}{dt}(x^2) + \frac{d}{dt}(y^2) = \frac{d}{dt}(13^2) \rightarrow 2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \rightarrow \frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}$$

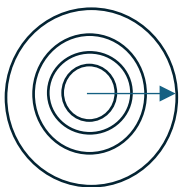
Finally, we plug in the values we know and solve: $\frac{dy}{dt} = -\frac{5}{12} \cdot 3 = -\frac{5}{4} \text{ ft/sec}$. **Don't forget units!**

Strategy:

- Draw a picture. What quantities are changing? Which are constants?
- Assign variables to these quantities (if not already given).
- Write your given information in terms of these quantities and their derivatives. Identify what you are looking for in terms of the variables.
- Write an equation that relates the variables and differentiate it with respect to time.
- Combine the given information with the equation and its derivative to find the unknown rate.

Next problem:

Problem 2: A pebble is dropped into a calm pond, causing ripples in the form of concentric circles. The radius r of the outer ripple is increasing at a constant rate of 1 foot per second. When the radius is 4 feet, at what rate is the total area A of the disturbed water changing?



If we draw a picture, we see that we have circular ripples where the radius is changing - and the area. Let r be the radius and A the area. Since these are circles, we know $A = \pi r^2$.

The problem gives us that $\frac{dr}{dt} = 1$ ft/s and we want to find $\frac{dA}{dt}$ when $r = 4$ ft.

We differentiate our equation: $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$. Substituting the given values, we get $\frac{dA}{dt} = 2\pi(4)(1) = 8\pi$ ft/s.

You will typically be provided with any underlying equations you may need, as above we used the Pythagorean Theorem, $x^2 + y^2 = z^2$ and area of a Circle, $A = \pi r^2$. Now, we will work some problems together. The solution to these problems, and other problems, are linked in Moodle.

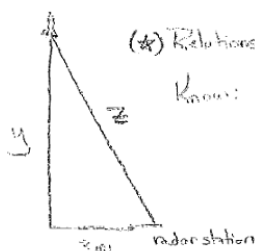
Partner up

Problem 3: A rocket that is launched vertically is tracked by a radar station located on the ground 3 miles from the launch site. What is the vertical speed of the rocket at the instant that its distance from the radar station is 5 mi and this distance is increasing at the rate of 5000 mi/h?

Problem 4: A spherical balloon is being inflated so that its volume is increasing at the rate of 5 cubic meters per minute. At what rate is the diameter increasing when the diameter is 12 m? (Remember that the volume of a sphere is $V = \frac{4}{3}\pi r^3$.)

Problem 6: The width of a rectangle is half its length. At what rate is its area increasing if its width is 10 cm and is increasing at 0.5 cm/s?

Problem 3: A rocket that is launched vertically is tracked by a radar station located on the ground 3 miles from the launch site. What is the vertical speed of the rocket at the instant that its distance from the radar station is 5 mi and this distance is increasing at the rate of 5000 mi/h?



(*) Relationship: $y^2 + 3^2 = z^2 \Rightarrow y^2 + 9 = z^2$

At the time in question,
Known: $\frac{dz}{dt} = 5000 \text{ mi/h}$ $z = 5 \text{ mi}$ want $\frac{dy}{dt}$

Derivative of (*) w/ respect to t :

$$2y \frac{dy}{dt} = 2z \frac{dz}{dt}$$

with $\frac{dy}{dt}$ unknown $\frac{dz}{dt}$ known

Let's re-evaluate (*), have $y^2 + 3^2 = 5^2$ at this time
so $y = 4$

Substituting:

$$2(4) \frac{dy}{dt} = 2(5) \frac{5000 \text{ mi/h}}{8}$$

$$\boxed{\frac{dy}{dt} = \frac{5000}{8} = 6250 \text{ mi/h}}$$

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(*) Relationship: $V = \frac{4}{3}\pi r^3$

← quantities changing

$$= \frac{4}{3}\pi \left(\frac{D}{2}\right)^3$$

$$= \frac{1}{6}\pi D^3$$

Need to relate r (radius) to D (diameter): $D = 2r$

Can substitute $r = \frac{D}{2}$

Need to know $\frac{dD}{dt}$ when $D = 12 \text{ m}$

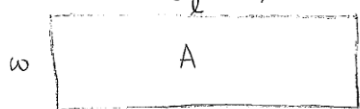
Known $\frac{dV}{dt} = 5 \text{ m}^3/\text{min}$

Derivative of (*): $\frac{dV}{dt} = \frac{3}{6}\pi D^2 \frac{dD}{dt} = \frac{1}{2}\pi D^2 \frac{dD}{dt}$
 ← Known → want to solve for

Substituting: $5 = \frac{1}{2}\pi (12)^2 \frac{dD}{dt}$

$$\Rightarrow \boxed{\frac{dD}{dt} = \frac{10}{144\pi} \approx 0.022 \text{ cm/min}}$$

Problem 6: The width of a rectangle is half its length. At what rate is its area increasing if its width is 10 cm and is increasing at 0.5 cm/s?



$w = \frac{1}{2}l$
or $l = 2w$
Want $\frac{dA}{dt}$

$A = wl = w(2w) = 2w^2$

↑
can substitute to eliminate a variable

when $w = 10 \text{ cm}$
 $\frac{dw}{dt} = 0.5 \text{ cm/s}$

Derivative of $A = 2w^2$:

$$\frac{dA}{dt} = 4w \frac{dw}{dt}$$

← solve for known

$$\boxed{\frac{dA}{dt} = 4(10)(0.5) = 20 \text{ cm}^2/\text{s}}$$